

Degree: B.Tech., Semester: Third-Course work
MID-SEMESTER EXAMINATION, September, 2024

Course Code: CMMTC08

Course Title: Mathematical Statistics

Duration: 1 hour 30 minutes

Max. Marks: 25

Note: Attempt all five questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO												
1(a)	If a random variable X takes the values 1, 2, 3, 4 such that $2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$, find the probability distribution of X .	2.5	CO1												
1(b)	Define Hypergeometric distribution and find its mean.	2.5	CO1												
2(a)	The mileage C in thousands of miles which car owners get with a certain kind of tyre is a random variable having the PDF $f(x) = \begin{cases} \frac{1}{20} e^{-\frac{x}{20}} & x > 0 \\ 0 & x \leq 0. \end{cases}$ Find the probability that one of these tyres will last i) at most 10,000 miles, ii) at least 30,000 miles.	2.5	CO1												
2(b)	An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. What is the probability that the first strike comes on the third well drilled?	2.5	CO1												
3(a)	Briefly explain the method of ordinates and the method of areas to fit a normal distribution for a given set of values.	2.5	CO1												
3(b)	Fit a Poisson's distribution to the following data and calculate theoretical frequencies: <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>Deaths</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Frequency</td><td>122</td><td>60</td><td>15</td><td>2</td><td>1</td></tr></table>	Deaths	0	1	2	3	4	Frequency	122	60	15	2	1	2.5	CO1
Deaths	0	1	2	3	4										
Frequency	122	60	15	2	1										
4(a)	Derive mean and variance of rectangular distribution.	2.5	CO2												
4(b)	A random variable X has a rectangular distribution over $(-3, 3)$, compute: (i) $P(X \leq 2)$ (ii) Find k for which $P(X > k) = 1/3$.	2.5	CO2												
5(a)	What is Sheppard's correction?	2.5	CO2												
5(b)	Does there exist a random variable X for which $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = 0.6$, where μ and σ are the mean and standard deviation of X .	2.5	CO2												